

The Calibration Catastrophe: nominally-90% CIs on self-duration

Bars = actual coverage of the agent's stated 90% CI containing τ_{wall} . Dashed green = the 90% target.

Dashed orange = the pre-registered P11 threshold of 0.5; only two cells cross it.

Every non-reasoning model under-covers by ≥ 40 percentage points; GPT-4o reaches 0%.

Model	Actual coverage of nominal 90% CI	cov CI width	width/actual	n
 Grounded target (90%)	90%	—	—	—
 Sonnet 4.6 + thinking	77%	13s	1.1x	30
 Kimi-K2.6	55%	14s	1.0x	29
 o4-mini (reasoning)	50%	4s	0.7x	30
 Sonnet 4.6	43%	25s	3.8x	30
 MiniMax-M2.7	33%	15s	1.6x	15
 Qwen3.6-27B	25%	11s	0.8x	12
 o3 (reasoning)	17%	50s	11.1x	29
 gpt-5.1	13%	49s	21.3x	30
 gpt-4o-mini	10%	20s	6.2x	30
 Haiku 4.5	7%	30s	8.3x	30
 gpt-4o	0%	20s	10.5x	30
P11 threshold 0.5				

Note: Sonnet 4.6 + thinking approaches the 90% target at 77% coverage, but its wider default interval (13 s against a 2–4 s baseline) reflects a broader duration vocabulary rather than chronoception. Both crossings rest on $n = 30$, under the $n \geq 100$ bar the pre-registered F4 sets for a falsification.